MOLES

A ***mole*** is used to measure, compare and calculate in Chemistry. It is defined as follows.

Mole: the number of atoms in 12.00 g of carbon-12.

The number of atoms in 1.00 mole of an element is given by Avogadro's Number.

Avogadro's Number: equal to 6.02×10^{23} particles/atoms/molecules/etc.

$\therefore$ 1.00 mole of Al atoms = 6.02×10^{23} atoms

1.00 mole of O_2 molecules = 6.02×10^{23} molecules

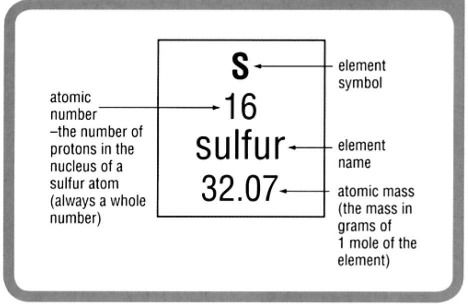
Molar Mass

A periodic table shows two important numbers when dealing with atoms. The ***atomic number*** is generally shown in the top left of the symbol box while the ***atomic mass*** of the element is shown elsewhere in the box.

All atomic masses of the Periodic Table are compared to C-12.

The periodic table provides you with important information about all of the chemical elements. One of these important pieces of information is the mass of one mole of the atoms in grams. This is known as the atomic mass.

Fig 4.5.3



Atomic Mass: the mass of an atom compared to $\frac{1}{12}$ of the mass of a carbon-12 atom.

Molecular/Formula Mass: the mass of a molecule (covalent substance) or formula unit (ionic substance) compared with $\frac{1}{12}$ of the mass of a carbon-12 atom.

Molar Mass: 1.00 mole of any substance is its atomic, molecular, or formula mass expressed in grams.

e.g. The molar mass is calculated by adding together the individual atomic masses.

$$\begin{aligned} M(CuSO_4) &= M(Cu) + M(S) + 4 \times M(O) \\ &= 63.5 + 32.1 + 4(16.0) \\ &= 159.6 \text{ gmol}^{-1}. \end{aligned}$$

The relationship between moles and mass is given by:

$$\boxed{\mathbf{n = \frac{m}{M}}}$$

where n = number of moles of a substance (mol)
m = mass of substance (g)
M = molar mass of the substance ($gmol^{-1}$)

Examples

1. What is the mass of 4.29 moles of $AlPO_4$?

$$n = \frac{m}{M} \qquad\qquad M(AlPO_4) = 122.0\ gmol^{-1}$$

$$\Rightarrow \quad 4.29 = \frac{m}{122.0}$$

$$\Rightarrow \quad m = 5.234 \times 10^2\ g$$

$$\therefore \quad \underline{\text{mass present} = 5.23 \times 10^2\ g}$$

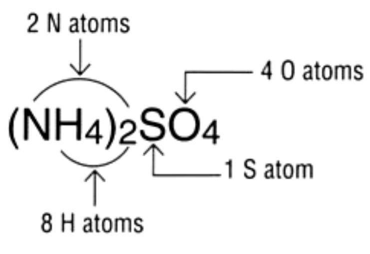
2.

What is the molar mass of ammonium sulfate?

Write out the correct formula: $(NH_4)_2SO_4$

Break it down into its elements.

$2N + 8H + 1S + 4O$



Work out the molar mass of the component elements and then sum them for the **molar mass of the compound**.

$= (2 \times 14.01) + (8 \times 1.008) + (32.06) + (4 \times 16.00)$

$= 28.02 + 8.06 + 32.06 + 64.00$

$= 132.14$ grams per mole

Molar masses can be calculated for any compound by breaking the formula down into its element components and adding them.

Fig 4.5.4

3.

Type example 1

Kate needs 0.25 moles of sucrose sugar ($C_6H_{12}O_6$) for a chemical reaction. How many grams will she need to mass out on the electronic balance?

$$m = ?$$

$$n(C_6H_{12}O_6) = 0.25$$

$$M(C_6H_{12}O_6) = (6 \times 12.01) + (12 \times 1.008) + (6 \times 16.00)$$

$$= 180.16$$

$$m = n \times M$$

$$= 0.25 \times 180.16 = 45.04 \text{ g}$$

Kate will need to mass out 45.04 g of sucrose.

Type example 2

A technician is cleaning up an acid spill and calculates that he needs 2.5 moles of calcium hydroxide, $Ca(OH)_2$, to neutralise the acid. What mass of calcium hydroxide needs to be massed out?

$$m = ?$$

$$n(Ca(OH)_2) = 2.5$$

$$M(Ca(OH)_2) = (1 \times 40.08) + (2 \times 16.00) + (2 \times 1.008)$$

$$= 74.10$$

$$m = n \times M$$

$$= 2.5 \times 74.10$$

$$= 185.24 \text{ g}$$

The technician will need to mass out 185.24 g of calcium hydroxide.

CHEMICAL CALCULATIONS AND EQUATIONS (STOICHIOMETRY)

A balanced chemical equation gives information about how the atoms rearrange during the chemical change process. It also gives information about the ***ratios between the number of particles of reactants and products.***

This allows calculations to be made and a prediction about the amounts of substances needed or produced during a chemical reaction. This is very important for controlling reactions and minimising costs in business and industry.

The key to chemical calculations is found at the mole level. It gives you the ***molar ratio*** of reactants to products, which is the key to solving problems in chemistry.

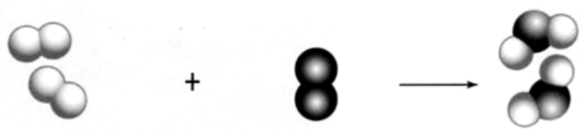
Word equation level	hydrogen gas + oxygen gas → water gas				
Chemical equation level	$2H_2\ (g) + O_2\ (g) \rightarrow 2H_2O\ (g)$				
Atomic level					
	2 hydrogen molecules	react with	oxygen molecule	to produce	2 water molecules
Multi-atomic level	2000 hydrogen molecules	react with	1000 oxygen molecules	to produce	2000 water molecules
Mole level	2 moles hydrogen molecules	react with	1 mole oxygen molecules	to produce	2 moles water molecules

Fig 4.5.6 Levels of understanding an equation for a reaction

Example

73.2 grams of $Al_2(CO_3)_3$ is reacted with excess HCl acid. Calculate:

(a) the mass of HCl used.
(b) the mass of CO_2 produced.

Step 1 Write a balanced equation.

$$Al_2(CO_3)_{3\,(s)} + 6\ HCl_{\,(aq)} \text{ ----> } 2\ AlCl_{3\,(aq)} + 3\ CO_{2\,(g)} + 3\ H_2O_{\,(l)}$$

Step 2 Put a tick under the substances you "know" the mass of, and a question mark under the substances you need to find.

$$\underset{\checkmark}{Al_2(CO_3)_{3\,(s)}} + \underset{?}{6\ HCl_{\,(aq)}} \text{ ----> } 2\ AlCl_{3\,(aq)} + \underset{?}{3\ CO_{2\,(g)}} + 3\ H_2O_{\,(l)}$$

Step 3 ***Convert the "known" mass to number of moles.***

$$n(Al_2(CO_3)_3) = \frac{m}{M} \qquad M(Al_2(CO_3)_3) = 234.6 \text{ gmol}^{-1}$$

$$= \frac{73.2}{234.6}$$

$$= 0.3120 \text{ mol}$$

Step 4 ***Use the balanced equation (mole ratios) to work out the number of moles of the "unknown" substance.***

$$n(HCl) = \frac{6}{1} \text{ x } n(Al_2(CO_3)_3)$$

$$= 6\ (0.3120)$$

$$= 1.872 \text{ mol}$$

Step 5 ***Convert the "unknown" number of moles to mass.***

(a)

$$n(HCl) = \frac{m}{M} \qquad M(HCl) = 36.5 \text{ gmol}^{-1}$$

$$\Rightarrow \quad 1.872 = \frac{m}{36.5}$$

$$\Rightarrow \quad m = 68.33 \text{ g}$$

$$\therefore \quad \underline{m(HCl) = 68.3 \text{ g}}$$

(b) ***In this calculation, repeat steps 4 and 5.***

$$n(CO_2) = \frac{3}{1} \text{ x } n(Al_2(CO_3)_3)$$

$$= 3\ (0.3120)$$

$$= 0.9360 \text{ mol}$$

$$n(CO_2) = \frac{m}{M} \qquad M(CO_2) = 44.0 \text{ gmol}^{-1}$$

$$\Rightarrow \quad 0.9360 = \frac{m}{44.0}$$

$$\Rightarrow \quad m = 41.18 \text{ g}$$

$$\therefore \quad \underline{m(CO_2) = 41.2 \text{ g}}$$